CSD 404 Internet of Things Lab

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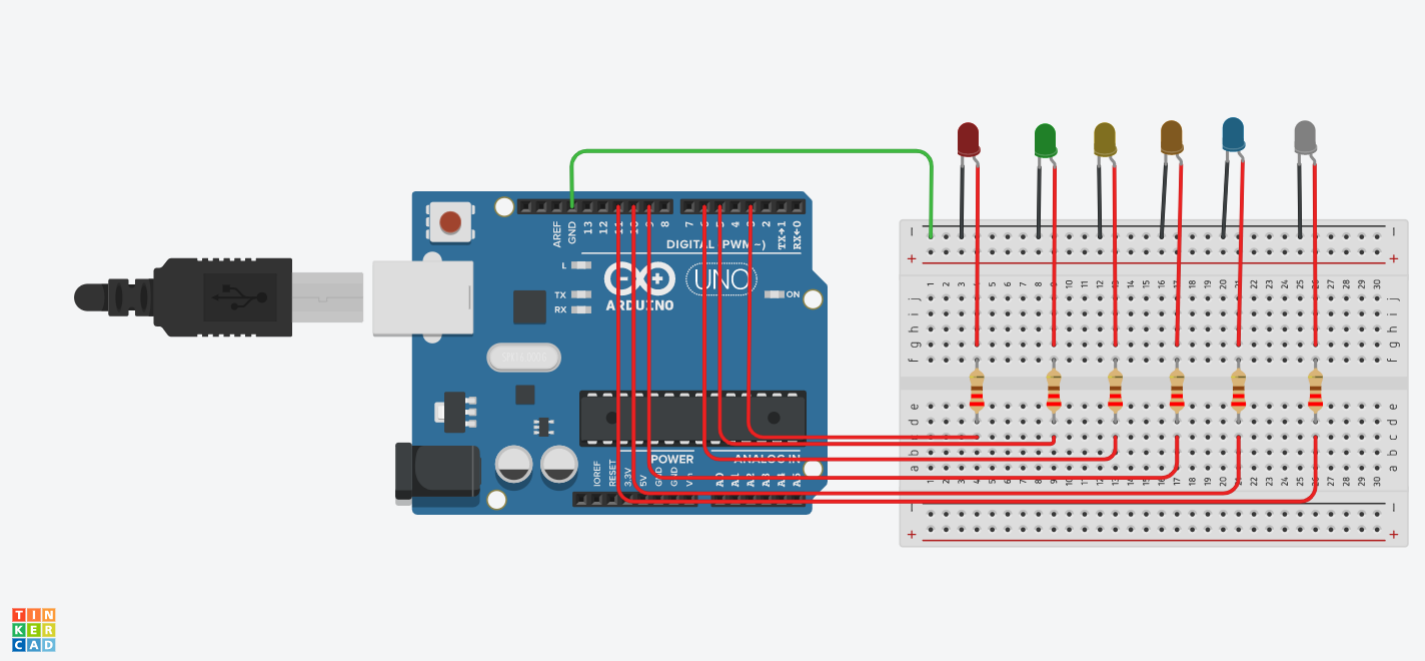
For this lab assignment, you will write basic Arduino programs.

Questions - Put the circuit diagram, code, and serial monitor output (Q3) after each question

1. Put **six** LEDs in a row successively; light up and dim them one after another, just like flowing water using PWM.

Circuit Diagram



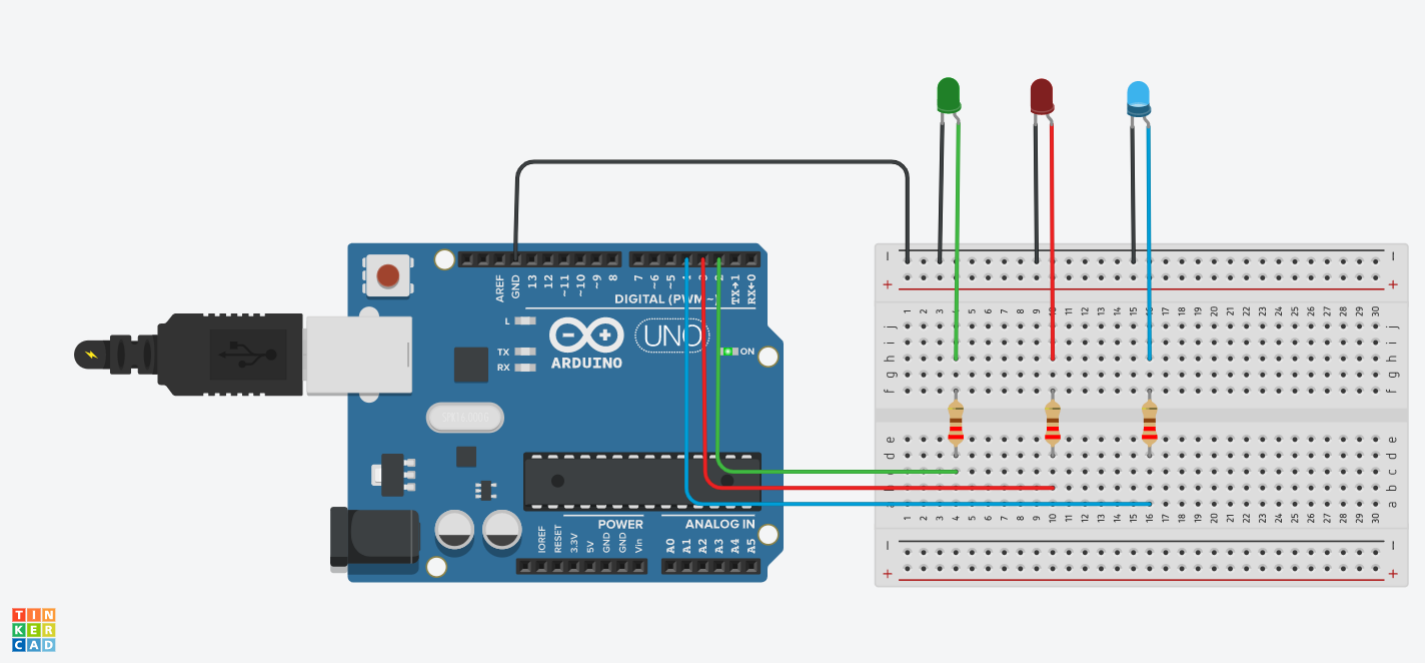

```
int brightness = 0;
int arr[6] = {3,5,6,9,10,11};
void setup()
{
  pinMode(arr[0], OUTPUT);
  pinMode(arr[1], OUTPUT);
  pinMode(arr[2], OUTPUT);
  pinMode(arr[3], OUTPUT);
  pinMode(arr[4], OUTPUT);
  pinMode(arr[5], OUTPUT);
}

void loop()
{
  int i=0;
  for(i=0; i<6; i++)
  {
    // Increae brightness
    for(brightness = 0; brightness <= 255; brightness+=5){
      analogWrite(arr[i], brightness);
      delay(10);
    }
    // Decrease brightness
    for(brightness = 255; brightness >= 0; brightness-=5){
      analogWrite(arr[i], brightness);
      delay(5);
    }
  }

}
```

2. Operate three LED's (Red, Green, and Blue) in such a manner that the Red must blink after a gap of one second, the Green must blink after a gap of half second, and the Blue must blink after a gap of two seconds.

Circuit Diagram




```
int DELAY_GREEN = 500;
int DELAY_BLUE = 2000;
int DELAY_RED = 1000;

void setup()
{
  pinMode(2, OUTPUT); // Green
  pinMode(3, OUTPUT); // Red
  pinMode(4, OUTPUT); // Blue
}

void loop()
{
  // Blink green
  digitalWrite(2, HIGH);
  digitalWrite(3, LOW);
  digitalWrite(4, LOW);
  delay(500);

  // Blink Red
  digitalWrite(3, HIGH);
  digitalWrite(2, LOW);
  digitalWrite(4, LOW);
  delay(1000);

  // Blink Blue
  digitalWrite(4, HIGH);
  digitalWrite(2, LOW);
  digitalWrite(3, LOW);
  delay(2000);
}
```

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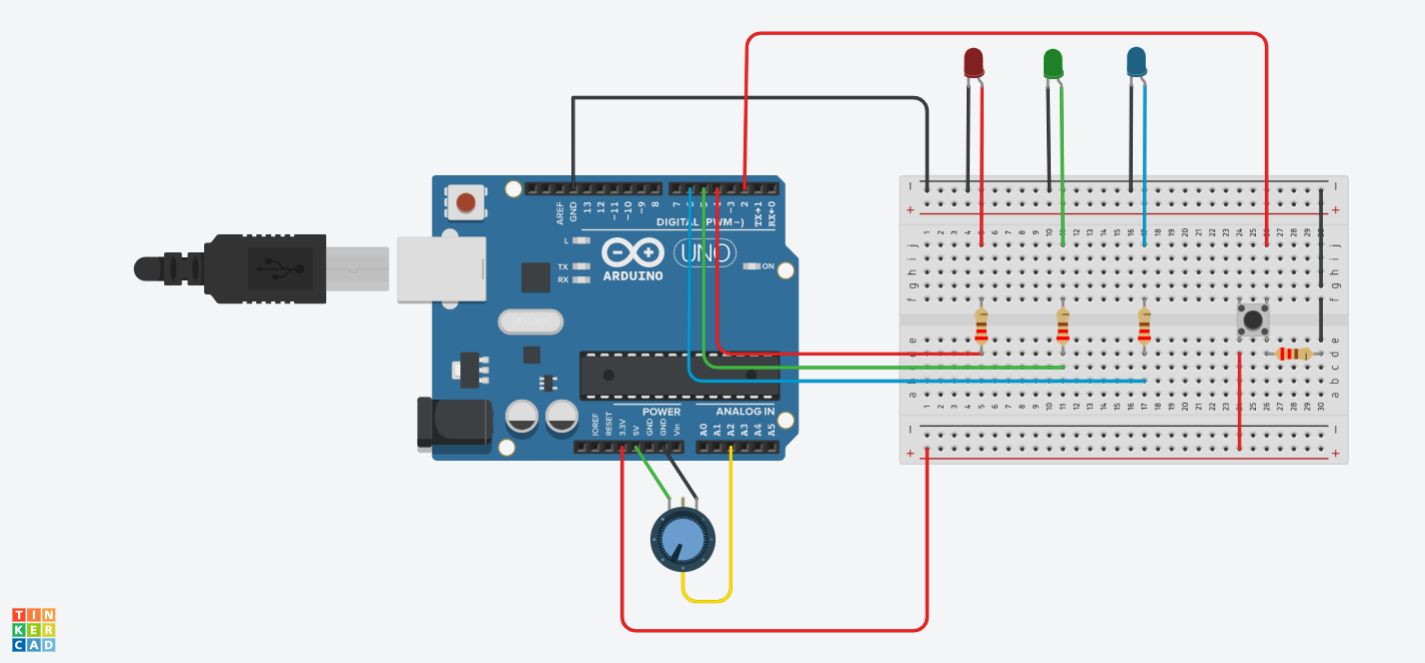
3. Using three LEDs (Red, Green, Blue), one Potentiometer, and one Push Button, write an Arduino sketch for displaying the tuned capacity of a potentiometer in percentage in the Serial Monitor. Use push button to activate or deactivate the LEDs. If tuned capacity is -

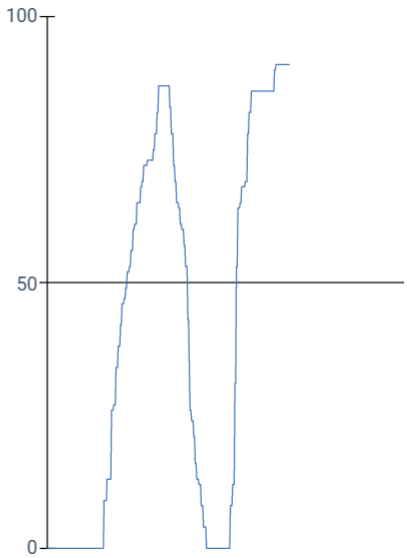
- < 40% then no LEDs will glow,
- >= 40% and < 55% then Red LED will glow,
- >= 55% and < 70% then Green LED will glow,
- >= 70% then Blue LED will glow.

Circuit Diagram




```
// Define constants
int RED = 4;
int GREEN = 5;
int BLUE = 6;
int pushButton = 2;
int sensorValue;
int potentiometerValue = 0;

void setup()
{
  pinMode(pushButton, INPUT);
  pinMode(RED, OUTPUT);
  pinMode(GREEN, OUTPUT);
  pinMode(BLUE, OUTPUT);
  pinMode(A2, INPUT);
  Serial.begin(9600);
}

void loop()
{
  potentiometerValue = analogRead(A2);
  int buttonState = digitalRead(pushButton);

  // Converting rotational degrees to output(mapping)
  sensorValue = (potentiometerValue/1023.0) * 100;
  Serial.println(sensorValue);

  // If button is pressed
  if (buttonState == 1)
  {
    // Cases for different percentage of rotations
    if (sensorValue < 40)
    {
      digitalWrite(GREEN, LOW);
      digitalWrite(BLUE, LOW);
      digitalWrite(RED, LOW);
    }
    else if(sensorValue >= 40 && sensorValue < 55)
    {
       digitalWrite(RED, HIGH);
    }
    else if(sensorValue >= 55 && sensorValue < 70)
    {
       digitalWrite(GREEN, HIGH);
    }
    else if(sensorValue >= 70)
    {
       digitalWrite(BLUE, HIGH);
    }

  }
  else
  {
      digitalWrite(GREEN, LOW);
      digitalWrite(BLUE, LOW);
      digitalWrite(RED, LOW);
  }
  delay(1);
}
```

Serial Monitor Image

Serial Monitor

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